

Extreme Spin Squeezing via Optimized One-Axis Twisting and Rotations

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The progress is shown for the generation of optimal squeezed states for extreme metrology. The scheme consists of an alternating series of one-axis twisting pulses and rotations, both of which are well-tailored to experimental constraints. The resulting states show a nontrivial gain proportional to the squeezing limit. We demonstrate that the squeezing using well-tailored one-axis twisting is comparable in the magnitude of the gain to implementing the standard twisting step when taking into account realistic limitations to phase twisting.

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1. INTRODUCTION

In quantum metrology, an ensemble of N noninteracting atoms provides a basic limit on the precision possible with $1/\sqrt{N}$, also known as the standard quantum limit (SQL). However, quantum entanglement also provides for the possibility of nonclassical correlations. The nonclassical correlations can provide an advantage in the scaling of the precision up to the Heisenberg limit (HL) with a gain of $\sqrt{N}$. This is a quantum advantage [1–3]. HL is reached in the limit of all-to-all coupling [4–6] that directly exploits the fundamental quantum nature of the measurement device. Approaches to achieving this precision include squeezing [7–10], hyperentanglement with a coherent multi-particle state, such as high-precision entangling is the effective reduction of applications that include the measurement [11] and at the fundamental level of precision [12,13] detection of pre-entangled states [14,15] (Hofmann [16], and recently [17]). Consequently, substantial effort has been devoted to design protocols that allow us to achieve the limit.

A well-known strategy to reach a scaling beyond the SQL is the utilization of spin-squeezed states (SSS) [18–24]. These states are characterized by nonclassical correlations that lower the variance of one measurement quadrature at the expense of increasing the variance of the quadrature orthogonal to the measured one. The standard proposal to generate SSS is based on squeezing

an initial state with one interaction Hamiltonian $\hat{H}_1$ and a second twisting step $\hat{H}_2$ (Hofmann [16]). The states that are generated by this twisting step are a scaling beyond the SQL in the limit of achieving the full Heisenberg limit. This is done with the idea of taking into account realistic limits in a typical experimental setup [17].

A relevant figure of merit for a spin-squeezed state is the squeezing gain, i.e., the additional phase sensitivity due to the measurement of the accumulated phase ϕ in a Ramsey interferometric measurement [18,25] relative to $\Delta\phi_{\text{SQL}} = 1/\sqrt{N}$. Then, an optimal SSS that saturates the optimal metrological advantage of the spin-squeezed state is achieved when the squeezing gain is approximately 3 times. The accumulation of metrological gain for the generation of a principle uncertainty compressed along one or two optical axes by Ramsey measurements. These optimal states result in the Heisenberg scaling [17,26]. They are also known as extreme SSSs (ESs) [27]. In this paper, we design a protocol to generate such extreme ESs.

Our proposed scheme uses only an experimentally demonstrated RMT interaction in an alternating manner with standard one-axis twisting. Thus, the advantage of our approach is that it can be realized in current experimental setups that employ RMT, specifically applications such as spectroscopy [15,28–30]. We find that with our specification of the RMT Hamiltonian we achieve a gain over Heisenberg scaling. Moreover, we can show that our approach reaches the scaling in a dispersive configuration of dispersive interactions in an optical setup as a means of achieving the effective RMT Hamiltonian [22].

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